

Nicholas Cabrera
Ethan Patterson

1. $T(n) = 3T(n/9) + \sqrt{n}$

$a=3$

$b=9$

$f(n) = \sqrt{n}$ or $n^{1/2}$

$d=1/2$

$a/b^d = (3/9^{1/2}) = 1$

This is the equal case!

$\log 3 / \log 9 = 1/2$

$T(n) = \theta(n^{1/2} \cdot \log n)$

2. $T(n) = 4T(n/2) + n^3$

$A = 4$

$B = 2$

$F(n) = n^3$

$D = 3$

$a/b^d = (4/2^3) = 4/8 = 1/2$ (less than 1)

This is the less than case!

$T(n) = \theta(n^3)$

3. $T(n) = 5T(n/4) + n \log n$

$A = 5$

$B = 4$

$f(n) = n \log n$

$D = 1$

$K = 1$

$a/b^d = (5/4^1) = 5/4$ (greater than 1)

This is the greater than case!

$\log 5 / \log 4 = \sim 1.16$

$T(n) = \theta(n^{\sim 1.16})$